

E3406416_1

The equation $x^4 + xy + x + y^3 + 4 = 0$ has no integer solutions

Size $h = 34$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + xy + x + y^3 + 4 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), with $(d_1, d_2) = (3, 6)$, and the fact that an odd prime p with $(-d_k/p) = -1$ divides $U_k^2 + d_k V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm forms $a^2 + db^2$

Throughout this section, d is a squarefree integer with $d \neq 0, -1$; we apply the results with $d \in \{3, 6\}$. For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-d/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + db^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -d(bc)^2 \equiv -d \pmod{p}$, so $-d$ is a square modulo p . As $(-d/p) \neq 0$, it is a non-zero square, so $(-d/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + db^2 \neq 0$, and $v_p(a^2 + db^2)$ is even for every odd prime p with $(-d/p) = -1$.*

Proof. If $d > 0$, then clearly $a^2 + db^2 > 0$. If $d < 0$, then $-d > 1$ is squarefree, hence not a square, so $a^2 + db^2 \neq 0$. Let p be an odd prime with $(-d/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + db'^2$ by Lemma 2. Since $a^2 + db^2 = p^{2k}(a'^2 + db'^2)$, we get $v_p(a^2 + db^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{4|d|}$, or, if $d < 0$, with $n' \equiv -n \pmod{4|d|}$. Then $(-d/n') = (-d/n)$.*

Proof. Since d is squarefree, we can write $-d = \varepsilon 2^e q$ with $\varepsilon \in \{1, -1\}$, $e \in \{0, 1\}$ and q odd and positive. By multiplicativity and the reciprocity laws,

$$\left(\frac{-d}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since $4|d|$ is divisible by $4q$, and by $8q$ if $e = 1$, the claim for $n' \equiv n$ follows. If $d < 0$ and $n' \equiv -n \pmod{4|d|}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. $\square$

Lemma 5. *Let n be a non-zero integer, and let s, u and m be positive integers such that every prime factor of s divides $2|d|$, both $2s$ and $4|d|s$ divide m , u is odd, $(-d/u) = -1$, and $n \equiv su \pmod{m}$. If $d > 0$, suppose also that $n > 0$. Then there is an odd prime p with $(-d/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by $4|d|$, n' is odd and $n' \equiv u \pmod{4|d|}$. If $d > 0$, then $n' > 0$, and if $d < 0$, then $|n'| \equiv \pm u \pmod{4|d|}$; in both cases Lemma 4 gives $(-d/|n'|) = (-d/u) = -1$. By multiplicativity, the product of $(-d/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(-d/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 2d$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

Lemma 6. *Let a, b, c be real numbers with $a > 0$, $c > 0$ and $256ac^3 > 27b^4$. Then $at^4 + bt + c > 0$ for every real number t .*

Proof. For $t = 0$ this is clear. For $t \neq 0$, the inequality between the arithmetic and geometric means of at^4 , $c/3$, $c/3$ and $c/3$ gives $at^4 + c \geq 4(at^4 c^3/27)^{1/4} = (256ac^3/27)^{1/4}|t| > |b||t| \geq -bt$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + xy + x + y^3 + 4$, so that (1) is the equation $F(x, y) = 0$. For the first identity, $d_1 = 3$, and we define

$$\begin{aligned} H_1 &= -y - 1, \\ T_1 &= x + y^2 - y + 1, \\ U_1 &= x^2, \\ V_1 &= 1, \\ G_1 &= -1, \\ A_1 &= 0, \\ B_1 &= 1, \end{aligned}$$

for the second identity, $d_2 = 6$, and we define

$$\begin{aligned} H_2 &= 4x - 2y + 3, \\ T_2 &= 32x^2 + 16xy + 56x + 8y^2 + 12y + 18, \\ U_2 &= 4x^2 + 16x + 8, \\ V_2 &= 3, \end{aligned}$$

$$\begin{aligned}
G_2 &= -16, \\
A_2 &= 0, \\
B_2 &= 1,
\end{aligned}$$

and, for the signs, we also use

$$\begin{aligned}
L_1 &= y + 1, \\
Q_1 &= y^3 + 4, \\
\Delta_1 &= 256y^9 + 3072y^6 - 27y^4 + 12180y^3 - 162y^2 - 108y + 16357, \\
Q_2 &= 16x^2 + 8xy + 28x + 4y^2 + 6y + 9, \\
M_2 &= 8x^4 + 64x^3 + 160x^2 + 128x + 59, \\
E_2 &= 192x^2 + 352x + 108.
\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 - 3V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - 6V_2^2 = G_2F, \quad (3)$$

$$A_1U_1 + B_1V_1 = 1, \quad (4)$$

$$A_2U_2 + B_2V_2 = 3, \quad (5)$$

$$F = x^4 + L_1x + Q_1, \quad (6)$$

$$\Delta_1 = 256Q_1^3 - 27L_1^4, \quad (7)$$

$$-8F = Q_2H_2 - M_2, \quad (8)$$

$$16Q_2 = (8x + 8y + 6)^2 + E_2. \quad (9)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_kT_k = U_k^2 + d_kV_k^2 \quad (k = 1, 2), \quad (10)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 72² residue pairs modulo 72, we find that (1) has exactly 36 solutions modulo 72, and that at each of them $H_1 \bmod 72$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 72$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-3/u) = -1$; here for $s = 1$, $2s = 2$ and $12s = 12$ divide 72. For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-6/u) = -1$; here for $s = 1$, $2s = 2$ and $24s = 24$ divide 72.

Step 2: signs. *The sign of H_1 .* Suppose that $H_1 \leq 0$. Since $H_1 = -y - 1$, this means $y \geq -1$. By (6), F is $x^4 + L_1x + Q_1$, where L_1 and Q_1 depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_1 and Δ_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_1 > 0$ and $\Delta_1 > 0$. For $y = -1$ and $y = 0$ we also have $Q_1 > 0$ and $\Delta_1 > 0$. Hence $Q_1 > 0$ and $\Delta_1 > 0$, so Lemma 6 with $a = 1$, $b = L_1$, $c = Q_1$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (8) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -4$, write

$x = -4 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $-3 \leq x \leq 0$. For $x \in \{-3, -2, 0\}$ we have $M_2 > 0$ and $E_2 > 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . Hence $M_2 > 0$ and $E_2 > 0$, so (9) gives $16Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

Step 3: primes p with $(-d_k/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(-d_k/p) = -1$, where $(d_1, d_2) = (3, 6)$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (10) and Lemma 3 show that $v_p(H_k T_k) = v_p(U_k^2 + d_k V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + d_k V_k^2$ by (10), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_k U_k + B_k V_k$. For $k = 1$, we have $R_1 = 1$. This is impossible. For $k = 2$, we have $R_2 = 3$. But p is odd, and $p \neq 3$ because $(-6/3) = 0$, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{72}$ with s and u as in Lemma 5 for $d = d_k$. By Step 2, $H_k \neq 0$, and $H_k > 0$ if $d_k > 0$. Hence Lemma 5 applies to $n = H_k$ and $d = d_k$, so there is an odd prime p with $(-d_k/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 72 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
5	17	29	41	53	65
11	23	35	47	59	71
$\mathcal{R}_2$, the residues of H_2					
13	37	61			